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DS-B

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HOMEWORK #10

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$$A=\begin{pmatrix}0&1&0\\0&0&1\\2&-5&4\end{pmatrix} \qquad A-\lambda I=\begin{pmatrix}-\lambda&1&0\\0&-\lambda&1\\2&-5&4-\lambda\end{pmatrix}$$

$$\det(A-\lambda I)=0=-\lambda\begin{vmatrix}-\lambda&1\\-5&4-\lambda\end{vmatrix}-1\begin{vmatrix}0&1\\2&4-\lambda\end{vmatrix}+0\begin{vmatrix}0&-\lambda\\2&-5\end{vmatrix}$$

$$-\lambda^3+4\lambda^2-5\lambda+2=0$$

$$\lambda_1=2,\quad \lambda_2=\lambda_3=1$$

For $\lambda=2$: $(A-2I)\vec{v}=\vec{0}$

$$\left[\begin{array}{ccc|c}-2&1&0&0\\0&-2&1&0\\2&-5&4-\lambda&0\end{array}\right]$$

$-2x_1+x_2=0$ — ①

$-2x_2+x_3=0$ — ②

$\boxed{x_3=x_3}$ — ③

$$\left[\begin{array}{ccc|c}-2&1&0&0\\0&-2&1&0\\0&-4&2&0\end{array}\right] R3+R1$$

$\boxed{x_2=2x_3/2}$

$-2x_1+\frac{x_3}{2}=0$

$\frac{-4x_1+x_3}{2}=0$

$$\sim\left[\begin{array}{ccc|c}\overset{x_1}{-2}&\overset{x_2}{1}&\overset{x_3}{0}&0\\0&-2&1&0\\0&0&0&0\end{array}\right] R3-2R1$$

$\boxed{x_1=\frac{x_3}{4}}$

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$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_3/4 \\ x_3/2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 1/4 \\ 1/2 \\ 1 \end{pmatrix}$$ Manual

Eigenvector : $\begin{pmatrix} 1/4 \\ 1/2 \\ 1 \end{pmatrix}$

For $\lambda = 1$:

$(A - \lambda)\vec{v} = \vec{0}$

$$\left[\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 2 & -5 & 3 & 0 \end{array}\right]$$

① $-x_1 + x_2 = 0$

② $-x_2 + x_3 = 0$

③ $x_3 = x_3$

$$\left[\begin{array}{ccc|c} \textcircled{-1} & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -3 & 3 & 0 \end{array}\right] R3 + 2R1$$

$x_2 = +x_3$

$-x_1 + x_3 = 0$

$x_1 = x_3$

$$\left[\begin{array}{ccc|c} \textcircled{-1} & 1 & 0 & 0 \\ 0 & \textcircled{-1} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] R3 - 3R1$$

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$$\vec{x} = \begin{pmatrix} \vec{x_1} \\ \vec{x_2} \\ \vec{x_3} \end{pmatrix} = \begin{pmatrix} x_3 \\ x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Linearly dependent

Eigenvector $= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

	AM	G.M
$\lambda = 1$	2	1
$\lambda = 2$	1	1
Σ	3	2

- The matrix is not diagonalizable as $\Sigma A.M \neq \Sigma G.M$

Q2: $A = \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}$ $\quad A - \lambda I = \begin{pmatrix} -1-\lambda & 0 & 1 \\ 3 & -\lambda & -3 \\ 1 & 0 & -1-\lambda \end{pmatrix}$

$$\det(A - \lambda I) = 0 = -1-\lambda \begin{vmatrix} -\lambda & -3 \\ 0 & -1-\lambda \end{vmatrix} + 0 \begin{vmatrix} 3 & -3 \\ 1 & -1-\lambda \end{vmatrix} + 1 \begin{vmatrix} 3 & -\lambda \\ 1 & 0 \end{vmatrix}$$

$$-\lambda^3 - 2\lambda = 0 \mapsto \lambda^2 = 0 \quad , \lambda_3 = -2$$

$\lambda_1 = 0, \lambda_2 = 0$, $\lambda_3 = -2$

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For $\lambda = 0$:

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 3 & 0 & -3 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R2+3R1 \\ R3+R1 \end{array}$$

$-x_1 + x_3 = 0$ $\quad x_3 = x_3$ $\quad x_2 = x_2$

$x_3 = x_1$

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_1 \end{pmatrix} = \begin{cases} \\ x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \\ \\ \end{cases}$$

Eigenvectors: $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

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For $\lambda = -2$

$$\begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix} - \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$\left[\begin{array}{ccc} 1 & 0 & 1 \\ 3 & 2 & -3 \\ 1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} \textcircled{1} & 0 & 1 & 0 \\ 0 & \textcircled{2} & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ R2-3R1 \\ R3-R1 \end{array}$$

$x_1 + x_3 = 0 \qquad 2x_2 - 6x_3 = 0 \qquad x_3 = x_3$

~~$x_1 + x_2$~~ $\qquad x_2 = 3x_3$

$x_3 = -x_1$

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -x_3 \\ 3x_3 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$$

$$\text{Eigenvector} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$$

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	A.M	G.M
$\lambda = 0$	2	2
$\lambda = -2$	1	1
Σ	3	3

As $\Sigma A.M = \Sigma G.M$, this matrix is diagonalizable.

(b) $A^{10}\begin{pmatrix}5\\1\\2\end{pmatrix} = (\lambda_1)^{10}\begin{pmatrix}0\\1\\0\end{pmatrix} + (\lambda_2)^{10}\begin{pmatrix}1\\0\\1\end{pmatrix} + (\lambda_3)^{10}\begin{pmatrix}-1\\3\\1\end{pmatrix}$

$= 0 + 0 + (-2)^{10}\begin{pmatrix}-1\\3\\1\end{pmatrix}$

$= \begin{pmatrix}-1024\\3072\\1024\end{pmatrix}$

(c) $A = PDP^{-1}$ where $P = \begin{bmatrix}0 & 1 & -1\\4 & 0 & 3\\0 & 1 & 1\end{bmatrix}$

$P^{-1} = \begin{bmatrix}3/2 & 1 & -3/2\\1/2 & 0 & 1/2\\-1/2 & 0 & 1/2\end{bmatrix}$

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$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Manual 2

$$A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3/2 & 1 & -3/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & 0 & 1/2 \end{bmatrix}$$

d) $A^{10} = PD^{10}P^{-1}$

$$= \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2^{10} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 3/2 & 1 & -3/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & 0 & 1/2 \end{bmatrix}$$

$$A^{10} = \begin{bmatrix} 0 & 0 & 0 \\ 1536 & 1024 & -1536 \\ 0 & 0 & 0 \end{bmatrix}$$

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